

Appendices

Appendix A

Some results from Calculus on $\mathbb{R}^n$

A.1 Results from Differentiation

Theorem A.1.1. (*Inverse Function Theorem*) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be continuously differentiable. Suppose that $\det Df(a) \neq 0$ for a given $a \in \mathbb{R}^n$. Then, there exists U and V open in $\mathbb{R}^n$ such that $a \in U$, $f(a) \in V$ and $f|_U : U \rightarrow V$ is a diffeomorphism.

Theorem A.1.2. (*Implicit Function Theorem*) Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be continuously differentiable in an open set of (a, b) and let $f(a, b) = 0$. Let M be the $m \times m$ matrix

$$D_{n+j}f^i(a, b); 1 \leq i, j \leq m$$

and suppose $\det M \neq 0$. Then there exists open sets $A \subset \mathbb{R}^n$, $B \subset \mathbb{R}^m$ with $a \in A$, $b \in B$ such that for all $x \in A$ there is a unique $g(x) \in B$ so that $f(x, g(x)) = 0$, and the g so obtained is differentiable.

Proof. Define $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n \times \mathbb{R}^m$ given by $F(x, y) := (x, f(x, y))$. This function is clearly differentiable, and moreover,

$$DF = \begin{pmatrix} \frac{\partial x^1}{\partial x^1} & \frac{\partial x^1}{\partial x^2} & \cdots & \frac{\partial x^1}{\partial x^n} & \frac{\partial x^1}{\partial y^1} & \frac{\partial x^1}{\partial y^2} & \cdots & \frac{\partial x^1}{\partial y^m} \\ \frac{\partial x^2}{\partial x^1} & \frac{\partial x^2}{\partial x^2} & \cdots & \frac{\partial x^2}{\partial x^n} & \frac{\partial x^2}{\partial y^1} & \frac{\partial x^2}{\partial y^2} & \cdots & \frac{\partial x^2}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial x^n}{\partial x^1} & \frac{\partial x^n}{\partial x^2} & \cdots & \frac{\partial x^n}{\partial x^n} & \frac{\partial x^n}{\partial y^1} & \frac{\partial x^n}{\partial y^2} & \cdots & \frac{\partial x^n}{\partial y^m} \\ \frac{\partial f^1}{\partial x^1} & \frac{\partial f^1}{\partial x^2} & \cdots & \frac{\partial f^1}{\partial x^n} & \frac{\partial f^1}{\partial y^1} & \frac{\partial f^1}{\partial y^2} & \cdots & \frac{\partial f^1}{\partial y^m} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x^1} & \frac{\partial f^m}{\partial x^2} & \cdots & \frac{\partial f^m}{\partial x^n} & \frac{\partial f^m}{\partial y^1} & \frac{\partial f^m}{\partial y^2} & \cdots & \frac{\partial f^m}{\partial y^m} \end{pmatrix}$$

And it is easy to see that $\det DF(a, b) = \det M \neq 0$. Therefore, from Inverse function theorem (with little modification), there exists an open set $A \times B$ of $\mathbb{R}^n \times \mathbb{R}^m$, and an open set $W \subset \mathbb{R}^n \times \mathbb{R}^m$ such that $(a, b) \in A \times B$, $(a, 0) \in W$ and $F : A \times B \rightarrow W$ is a diffeomorphism. $F^{-1} : W \rightarrow A \times B$ is also a diffeomorphism. Moreover, note that $F^{-1}(p, q)$ is of the form $(p, k(p, q))$ for some differentiable function $k : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$. $\pi \circ F(x, y) = \pi \circ (x, f(x, y)) = f(x, y)$ so $\pi \circ F \equiv f$. Look at $f \circ (p, k(p, q)) = f \circ F^{-1}(p, q) = \pi(p, q) = q$ so $f \circ (p, k(p, 0)) = 0$. Our $g(p)$ is given by $k(p, 0)$, and it is differentiable as a composition of differentiable functions.

■

Theorem A.1.3. (A corollary of A.1.2) *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ ($p \leq n$) be continuously differentiable in an open set containing $a \in \mathbb{R}^n$. If $f(a) = 0$ and the matrix $D_j(f^i(a))_{p \times n}$ has rank p , then there exists an open neighbourhood A of a and a diffeomorphism $h : A \rightarrow h(A) \subset \mathbb{R}^n$ such that $f \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$.*

Proof. Consider $f : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^p$. Suppose that the matrix $Q = D_{n-p+i}(f^j); i, j \leq p$ be invertible ($\det Q \neq 0$). Let, as before, $F : \mathbb{R}^{n-p} \times \mathbb{R}^p \rightarrow \mathbb{R}^{n-p} \times \mathbb{R}^p$ given by

$$F(p, q) = (p, f(p, q))$$

Since $\det DF \neq 0$, there exists, around $a = (a^1, a^2, \dots, a^{n-p}, a^{n-p+1}, \dots, a^n)$ a neighbourhood $A_{n-p} \times A_p \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ and a neighbourhood $W \subset \mathbb{R}^{n-p} \times \mathbb{R}^p$ containing $F(a) = (a_1, a_2, \dots, a_{n-p}, 0)$ such that $F : A_{n-p} \times A_p \rightarrow W$ is invertible and whose inverse is continuously differentiable. Note that, the inverse of F ought to be of the form $F^{-1}(x_1, x_2, \dots, x_{n-p}, y_{n-p+1}, \dots, y_n) = (\vec{x}, b(\vec{x}, \vec{y}))$ (let $\vec{x} = (x_1, x_2, \dots, x_{n-p}) \in \mathbb{R}^{n-p}$ and $\vec{y} = (y_{n-p+1}, y_{n-p+2}, \dots, y_n) \in \mathbb{R}^p$) where it is understood that $b : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is continuously differentiable. Also note that $\pi_2 \circ F \equiv f$. We have that

$$f \circ F^{-1}(x_1, x_2, \dots, x_{n-p}, x_{n-p+1}, \dots, x_n) =$$

$$\pi_2(x_1, x_2, \dots, x_{n-p}, x_{n-p+1}, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$$

Case 2: If the indices where the matrix $\{(\partial_i f^j(a)); 1 \leq j \leq p, i = i_1, i_2, \dots, i_p\}$ is not the last p indices, we write the linear transformation $\text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that takes these p indices to the last p indices. This

is a linear transformation that is invertible since it takes basis elements to basis elements. If we look then, at $f \circ \text{Swap} : \mathbb{R}^n \rightarrow \mathbb{R}^p$, we see that this is a function of the form of Case 1, so we have a h such that $f \circ \text{Swap} \circ h(x_1, x_2, \dots, x_n) = (x_{n-p+1}, x_{n-p+2}, \dots, x_n)$ which means our desired function is $\text{Swap} \circ h$.

■

A.2 Results from Integration

A.2.1 The Riemann Integral

Definition A.2.1. (Partition) For closed interval $[a, b]$ in $\mathbb{R}$, a partition is simply a collection of sub-intervals $[a_i, a_{i+1}]$ so that $a_0 = a < a_1 < a_2 < \dots < a_k = b$.

For a closed rectangle R in $\mathbb{R}^n$, a partition P , usually written along with its subpartitions (as in, $P \equiv (P_1, P_2, \dots, P_n)$), is a collection of subrectangles R_i (of a particular form) so that $R = \cup_i R_i$ and $R_i^\circ \cap R_j^\circ = \emptyset$. Explicitly, if

$$R = [a^1, b^1] \times [a^2, b^2] \times \dots \times [a^n, b^n]$$

,

$$P_i \equiv [a^i = a_0^i < a_1^i < a_2^i \dots a_{k_i}^i = b^i]$$

and I_j^i denotes the j -th subinterval of the i -th partition, the collection of all possible sub rectangles of R of the form $\{I_{j_1}^1 \times I_{j_2}^2 \times \dots \times I_{j_n}^n : 1 \leq j_q \leq \text{number of intervals}(P_q)\}$ is a partition of R .

Definition A.2.2. (Upper and Lower sums) Let R be a rectangle in $\mathbb{R}^n$, $f : R \rightarrow \mathbb{R}$ be a bounded function and let P be a partition of R .

$$\mathcal{L}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\inf_{x \in S} f(x) \right) |S|$$

$$\mathcal{U}(P, f) := \sum_{S \in P(S \text{ is a subrectangle})} \left(\sup_{x \in S} f(x) \right) |S|$$

where $|S|$ is the volume of the subrectangle S . For convenience, we denote $\inf_{x \in S} f(x)$ by $m_S(f)$ and $\sup_{x \in S} f(x) = M_S(f)$.

Lemma A.2.3. For a bounded function $f : R \rightarrow \mathbb{R}$,

$$\mathcal{L}(P, f) \leq \mathcal{U}(P, f)$$

for a given partition P .

Proof. If S_i are subrectangles of S coming from P' , then $m_S(f)|S| = m_S(f)(|S_1| + |S_2| + \cdots + |S_i|) \leq \sum_{S_i \in S} m_{S_i}(f)|S_i|$. This proves the result for lower sums, and the proof for upper sums is similar. ■

Lemma A.2.4. If P' is a refinement of P (i.e. each subrectangle of P' is contained in some subrectangle of P) then

$$\mathcal{L}(P, f) \leq \mathcal{L}(P', f)$$

and

$$\mathcal{U}(P', f) \leq \mathcal{U}(P, f)$$

Lemma A.2.5. If P and P' are any two partitions and $f : R \rightarrow \mathbb{R}$ a bounded function, then $\mathcal{L}(P', f) \leq \mathcal{U}(P, f)$.

Proof. Let P'' be a partition of both P and P' (i.e. the subrectangles of P'' are contained in some subrectangle of P as well as some subrectangle of P'). Then by the previous lemma,

$$\mathcal{L}(P', f) \leq \mathcal{L}(P'', f) \leq \mathcal{U}(P'', f) \leq \mathcal{U}(P, f)$$

Now we are in a position to define the Riemann Integral. ■

Definition A.2.6. ($\int_R f$) Let $f : R \rightarrow \mathbb{R}$ be a bounded function from a rectangle R . We say f is *Riemann integrable* if

$$\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f)$$

where $\sup$ and $\inf$ are taken over all partitions P of R . In that case we say $\sup_P \mathcal{L}(P, f) = \inf_P \mathcal{U}(P, f) := \int_R f$.

Lemma A.2.7. A bounded function $f : R \rightarrow \mathbb{R}^n$ is Riemann Integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that $\mathcal{U}(P, f) - \mathcal{L}(P, f) < \varepsilon$.

A.2.2 Measure Zero and Content Zero

Definition A.2.8. (Measure Zero) A set $A \subset \mathbb{R}^n$ is said to be *measure zero* if for every $\varepsilon > 0$, there exists a countable collection of closed rectangles R_i such that $A \subset \bigcup_i R_i$ and $\sum_i |R_i| < \varepsilon$.

Lemma A.2.9.

1. *Subsets of Measure Zero sets are Measure Zero.*
2. *Countable union of Measure Zero sets is Measure Zero.*

Proof. (1) Is trivial.

For (2), consider E_i a countable collection of measure zero sets. Let $\varepsilon > 0$. Consider $\varepsilon/2^i$. There exists a collection of closed rectangles $\{R_{ij} : j \in \mathbb{N}\}$ such that $E_i \subseteq \bigcup_j R_{ij}$ and $\sum_j |R_{ij}| < \varepsilon/2^i$. Look at the collection $\{R_{ij} : i, j \in \mathbb{N}\}$ as a whole. This is a countable collection of closed rectangles that cover $\bigcup_i E_i$, and note that $\sum_{i,j} |R_{i,j}| < \varepsilon$.

■

Definition A.2.10. (Content Zero) A set $A \subset \mathbb{R}^n$ is said to have *content zero* if for every $\varepsilon > 0$, there exists a *finite* collection of closed rectangles R_i such that $A \subseteq \bigcup_{i=1}^n R_i$ and $\sum_{i=1}^n |R_i| < \varepsilon$.

Lemma A.2.11. *If A is compact and has measure zero, then it has content zero.*

A.2.3 Integrable Functions

Definition A.2.12. (Oscillation) Let $f : X \rightarrow \mathbb{R}$ be a bounded function, where X is a metric space. Define

$$M(f, a, \delta) := \sup_{x \in B_\delta(a)} f(x)$$

and

$$m(f, a, \delta) := \inf_{x \in B_\delta(a)} f(x)$$

Let

$$\text{osc}(f, a, \delta) := M(f, a, \delta) - m(f, a, \delta)$$

Note first, that $\text{osc}(f, a, \delta)$ is decreasing for decreasing δ . Now we define

$$\boxed{\text{osc}(f, a) := \lim_{\delta \rightarrow 0} \text{osc}(f, a, \delta)}$$

An alternative characterisation of oscillation is the following:

Lemma A.2.13. *Let f be as in definition A.2.12. The quantity $\sup_{x,y \in B_\delta(a)} |f(y) - f(x)|$ is decreasing for decreasing δ and*

$$\text{osc}(f, a) = \lim_{\delta \rightarrow 0} \left(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| \right) = \inf_{\delta > 0} \left(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| \right)$$

Lemma A.2.14. *$f : X \rightarrow \mathbb{R}$ (where X is a metric space) is continuous at a if and only if $\text{osc}(f, a) = 0$*

Proof. $\implies$) Let f be continuous at a . For any given $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in B_\delta(a)$ we have

$$|f(x) - f(a)| < \varepsilon/2 \text{ and } |f(y) - f(a)| < \varepsilon/2$$

Adding these we get

$$|f(x) - f(y)| < \varepsilon$$

for $x, y \in B_{\delta(\varepsilon)}(a)$. Therefore, $\sup_{x,y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$. Since for every $\varepsilon > 0$, there exists some $\delta(\varepsilon) > 0$ such that $\sup_{x,y \in B_{\delta(\varepsilon)}(a)} |f(x) - f(y)| < \varepsilon$, we have $\text{osc}(f, a) = 0$.

$\impliedby$) Let $\text{osc}(f, a) = \inf_{\delta > 0} \left(\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| \right) = 0$. This means that for every $\varepsilon > 0$, there exists $\delta > 0$ such that $\sup_{x,y \in B_\delta(a)} |f(x) - f(y)| < \varepsilon$, which means that

$$|f(x) - f(a)| < \varepsilon$$

whenever $x \in B_\delta(a)$.

■

Lemma A.2.15. (Recharacterisation of oscillation in terms of rectangles in $\mathbb{R}^n$)

$$\text{osc}(f, a) = \inf_{\delta > 0} \left(\sup_{x \in R_\delta(a)} f(x) - \inf_{x \in R_\delta(a)} f(x) \right)$$

Where $R_\delta(a)$ is a cube centred at a with side length δ .

Proof. Note that

$$B_{\delta/\sqrt{n}}(a) \subseteq R_\delta(a) \subseteq B_{\delta\sqrt{n}}(a)$$

From here the proof is trivial.

Lemma A.2.16. *Let A be a closed rectangle and $f : A \rightarrow \mathbb{R}$ be a bounded function such that $\text{osc}(f, a) < \varepsilon$ for every $a \in A$. Then there exists a partition P of A such that $\mathcal{U}(f, P) - \mathcal{L}(f, P) < \varepsilon \cdot |A|$*

Proof. We are given a fixed $\varepsilon_0 > 0$. We have that $\text{osc}(f, a) < \varepsilon_0$ for all $a \in A$. $\inf_{\delta > 0} (\sup_{x \in R_\delta(a)} f(x) - \inf_{y \in R_\delta(a)} f(y)) < \varepsilon_0$ for every $a \in A$. This means that, for every $a \in A$, there exists $\delta_a > 0$ such that $\sup_{x \in R_{\delta_a}(a)} f(x) - \inf_{x \in R_{\delta_a}(a)} f(x) < \varepsilon_0$. Collect all such $R_{\delta_a}(a)$. This is a cover of A , for which a finite subcover $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$ exists. Consider the partition P of A formed by extending the grids of all the cubes $\{R_{\delta_{a_j}}(a_j) : j \leq N\}$. The final partition P 's subrectangles J_i have the property that

$$\sup_{x \in J_i} f(x) - \inf_{x \in J_i} f(x) < \varepsilon_0$$

Essentially,

$$M_S(f) - m_S(f) < \varepsilon_0$$

for S (a subrectangle) $\in P$. This gives us that $\mathcal{U}(f, P) - \mathcal{L}(f, P) = \sum_{S \in P} (M_S(f) - m_S(f))|S| < \varepsilon_0 \sum_{S \in P} |S| = \varepsilon_0 |A|$

Theorem A.2.17. *Let A be a closed rectangle and $f : A \rightarrow \mathbb{R}$ be a bounded function. Let $B := \{x \in A : f \text{ not continuous at } x\}$. Then f is integrable if and only if B is a set of measure 0.*

Proof. Note that $B := \{x \in A : \text{osc}(f, x) > 0\}$ which could be written as $\cup_{n \in \mathbb{N}} \{x \in A : \text{osc}(f, x) \geq 1/n\}$, where we note that $\{x \in A : \text{osc}(f, x) \geq t\}$ is closed for any $t > 0$ (If $\text{osc}(f, a) < \varepsilon$ then there would exist a ball around a such that $\text{osc}(f, x) < \varepsilon$ for x in said ball).

$\implies$) Let B be measure 0. Consider the closed (and hence compact) set $B_\varepsilon := \{x \in A : \text{osc}(f, x) \geq \varepsilon\}$ whose measure (since $B_\varepsilon \subseteq B$) is 0. We thus have a finite collection of rectangles R_i that cover B_ε and $\sum_{i=1}^N |R_i| < \varepsilon$. Consider the partition P of A generated by extending the grid formed by R_i and collecting all subrectangles K_i . These can be classified into two subcategories (which together partition A):

$$1. (\mathbb{S})^1 := \{K_i : K_i \cap B_\varepsilon \neq \emptyset\}$$

$$2. (\mathbb{S})^2 := \{K_i : K_i \cap B_\varepsilon = \emptyset\}$$

We now look at $\mathcal{U}(f, P) - \mathcal{L}(f, P) = \sum_{S \in (\mathbb{S})^1 \cup (\mathbb{S})^2} (M_S(f) - m_S(f))|S|$.

Look at the first sum:

$$\sum_{S \in (\mathbb{S})^1} (M_S(f) - m_S(f))|S|$$

Since f is bounded, $|f(x)| \leq M$ and hence $\sum_{S \in (\mathbb{S})^1} (M_S(f) - m_S(f))|S| \leq 2M \sum_{S \in (\mathbb{S})^1} |S|$. Since we know that S_1 consists of the subrectangles of the cover elements (and only the cover elements) we note that $\sum_{S \in (\mathbb{S})^1} |S| \leq \varepsilon$. Hence we have the first sum $\leq 2M\varepsilon$

Look at the second sum:

$$\sum_{S \in (\mathbb{S})^2} (M_S(f) - m_S(f))|S|$$

For $S \in (\mathbb{S})^2$, we have $\text{osc}(f, x) < \varepsilon$. By A.2.16 we have that there exists a partition of S , call it P_S such that $\mathcal{U}(f, P_S) - \mathcal{L}(f, P_S) \leq \varepsilon|S|$. Extending the grid for each of these subrectangles in P_S forms a new partition P' which is a refinement of P as a whole, such that $\sum_{S' \subset S; S' \in P'} (M_{S'}(f) - m_{S'}(f))|S'| \leq \varepsilon|S|$, so as a whole,

$$\begin{aligned} \mathcal{U}(f, P') - \mathcal{L}(f, P') &= \sum_{S' \in (\mathbb{S})^1} \sum_{S'' \subset S'; S'' \in P'} (M_{S''}(f) - m_{S''}(f))|S''| + \\ &\quad \sum_{S' \in (\mathbb{S})^2} \sum_{S'' \subset S'; S'' \in P'} (M_{S''}(f) - m_{S''}(f))|S''| \\ &\leq 2M\varepsilon + \sum_{S' \in (\mathbb{S})^2} \varepsilon|S'| \leq 2M\varepsilon + \varepsilon|A| \end{aligned}$$

Hence f is integrable by criteria A.2.7 we have that f is integrable.

$\Leftarrow$) Let f be integrable. The set of discontinuities B is $\cup_{n \in \mathbb{N}} B_n$ where $B_n := \{x \in A : \text{osc}(f, x) \geq 1/n\}$ (closed and hence compact). For $a \in B_n$,

we have $\text{osc}(f, a) \geq 1/n$ which means that $\inf_{\delta>0}(M_{R_\delta}(f) - m_{R_\delta}(f)) \geq 1/n$. Let P be the partition of A such that

$$\mathcal{U}(f, P) - \mathcal{L}(f, P) \leq \varepsilon/n$$

and let SS be the collection of subrectangles of P which intersect with B_n . These form a cover for B_n . Therefore, for S such that $S \cap B_n \neq \emptyset$, we have $\exists x \in S$ such that $\inf_{\delta>0}(M_{R_\delta}(f) - m_{R_\delta}(f)) \geq 1/n$ which implies $M_S(f) - m_S(f) \geq 1/n$ and hence

$$\varepsilon/n \geq \sum_{S \in SS} (M_S(f) - m_S(f))|S| \geq 1/n \sum_{S \in SS} |S|$$

which gives us that $\sum_{S \in SS} |S| \leq \varepsilon$. Therefore, B_n is content zero and hence B is of measure zero. ■

Definition A.2.18. ($\int_C f$ for arbitrary set C) Let A be any rectangle containing C (in the case where we can find one). Then we define

$$\int_C f := \int_A f \cdot \chi_C$$

where $\chi_C(x) = \begin{cases} 1 & \text{if } x \in C \\ 0 & \text{if } x \notin C \end{cases}$

Theorem A.2.19. Function $\chi_C : A \rightarrow \mathbb{R}$ is integrable if and only if ∂C is a set of measure zero.

Proof. Note that, for $x \in C^\circ$, χ_C is continuous. And likewise for $x \in (\mathbb{R}^n - C)^\circ$. The only place where χ_C is discontinuous is precisely on ∂C . The result follows from [A.2.17](#). ■

Definition A.2.20. (Jordan-Measurable) A bounded set C whose boundary has measure zero is called **Jordan Measurable**.

Theorem A.2.21. If $f : A \rightarrow \mathbb{R}$ is a non-negative function and $\int_A f = 0$, then $B = \{x : f(x) > 0\}$ is a set of measure zero.

Proof. Let $B_n := \{x \in A : f(x) \geq 1/n\}$ and note that $B = \cup_n B_n$. Since f is said to be integrable, we note that $\inf_P \mathcal{U}(f, P)$ must be equal to the integral $\int_A f = 0$. Let $\varepsilon > 0$ be arbitrary and consider that partition P such that $\mathcal{U}(P, f) < \varepsilon/n$. Look at $\mathbb{J} := \{S \in P : S \cap B_n \neq \emptyset\}$. This covers B_n . Moreover, $1/n \sum_{S \in \mathbb{J}} |S| \leq \sum_{S \in \mathbb{J}} M_S(f) |S| \leq \mathcal{U}(f, P) \leq \varepsilon/n$ (Since $f(x) \geq 1/n$ for elements in B_n , and each S has some intersection with B_n). This implies $\sum_{S \in \mathbb{J}} |S| \leq \varepsilon$ and $\cup_{S \in \mathbb{J}} S$ covers B_n . Therefore, B_n is content zero set, and hence B is a measure zero set. ■

A.2.4 Fubini's Theorem

Definition A.2.22. (Lower and Upper Integrals) If $f : A \rightarrow \mathbb{R}$ is a bounded function on a closed rectangle $A \subset \mathbb{R}^n$, then regardless of whether f is integrable, the least upper bound of the lower sums and the greatest lower bound of the upper sums both exist. They are called **Lower** and **Upper integrals** of f on A , denoted by:

$$\mathbf{L} \int_A f$$

and

$$\mathbf{U} \int_A f$$

Theorem A.2.23. (Fubini) Let $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^m$ be closed rectangles and let $f : A \times B \rightarrow \mathbb{R}$ be integrable. For $x \in A$, let $g_x : B \rightarrow \mathbb{R}$ be given by $g_x(y) = f(x, y)$ (and likewise define $h_y : A \rightarrow \mathbb{R}$ as $h_y(x) = f(x, y)$). Let

$$\mathcal{L}(x) := \mathbf{L} \int_B g_x = \mathbf{L} \int_B f(x, y) dy$$

and

$$\mathcal{U}(x) := \mathbf{U} \int_B g_x = \mathbf{U} \int_B f(x, y) dy$$

then $\mathcal{L}(x) : A \rightarrow \mathbb{R}$ and $\mathcal{U}(x) : A \rightarrow \mathbb{R}$ are both integrable on A and

$$\begin{aligned} \int_{A \times B} f &= \int_A \mathcal{L}(x) dx = \int_A \left(\mathbf{L} \int_B f(x, y) dy \right) dx \\ \int_{A \times B} f &= \int_A \mathcal{U}(x) dx = \int_A \left(\mathbf{U} \int_B f(x, y) dy \right) dx \end{aligned}$$

Proof. (sketch)

■

A.3 Partitions of Unity

Lemma A.3.1. *There exists a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which is positive on $(-1, 1)$, 0 elsewhere and is C^∞*

Proof. Consider

$$f := \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & \text{if } x \in (-1, 1) \\ 0 & \text{elsewhere} \end{cases}$$

This function is C^∞ and is positive on $(-1, 1)$ and 0 elsewhere.

■

Lemma A.3.2. *Let $a \in \mathbb{R}^n$, $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by*

$$g(x) = f([x^1 - a^1]/\varepsilon) f([x^2 - a^2]/\varepsilon) \cdots f([x^n - a^n]/\varepsilon)$$

g is then C^∞ , and has positive values only on $(a^1 - \varepsilon, a^1 + \varepsilon) \times (a^2 - \varepsilon, a^2 + \varepsilon) \cdots \times (a^n - \varepsilon, a^n + \varepsilon)$.

Appendix B

Exterior Forms

B.1 Definitions and Results

Definition B.1.1. (Exterior Forms) Let $k \geq 1$ and $n \in \mathbb{N}$. A k -form on $\mathbb{R}^n$ is a map

$$f : \underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \cdots \times \mathbb{R}^n}_{k \text{ copies}} \rightarrow \mathbb{R}$$

such that

1. f is linear in each of its k coordinates.
2. For every $\sigma \in \text{sym}(k)$, we have $f(x_{\sigma(1)}, x_{\sigma(2)} \cdots x_{\sigma(k)}) = \text{sgn}(\sigma)f(x_1, x_2, \cdots x_k)$.

Definition B.1.2. ($\Lambda^k(\mathbb{R}^n)$) We denote the space of all k -forms by $\Lambda^k(\mathbb{R}^n)$. By convention, we set

$$\Lambda^0(\mathbb{R}^n) = \mathbb{R}$$